

PUSHPA.

Predictive Unified System for Forest Protection & Anti-Smuggling

MISSION BRIEF & EXECUTIVE DOSSIER

A comprehensive technical architectural blueprint detailing multi-spectral Earth Observation (Copernicus Sentinel-2), automated NDVI bi-temporal vegetation loss vectorization, real-time vehicle GPS telemetry corridor anomaly detection, permit registry verification, and Explainable AI (XAI) risk scoring — augmented with a next-generation **Real-Time Smuggling Prevention Architecture** featuring Arboreal Acoustic IoT Sensor Grids, Autonomous Interceptor UAVs, Smart Weighbridge ANPR Checkpoints, and Blockchain Timber Chain-of-Custody.

TARGET BIOME & DOMAIN

Western Ghats & Tropical Reserves

CORE TECHNOLOGIES

FastAPI, Sentinel-2, React, Leaflet, Three.js

SECURITY & VERIFICATION

Explainable AI (0-100 Risk Engine)

CLASSIFICATION

Tactical Intelligence Whitepaper

VERSION & RELEASE

v2.4.0 Production Specification

JURISDICTION MODE

Human-in-the-Loop Advisory

1. Executive Summary & Core Problem Statement

Forests across critical ecological hotspots such as the Nilgiri Biosphere Reserve, Anamalai Range, and Periyar Corridor face unprecedented threats from illegal logging, organized timber poaching (e.g., Red Sanders, Rosewood, and Teak), and clandestine transit smuggling. Traditional forest protection methods rely heavily on manual foot patrols, delayed physical reports, and isolated boundary checkpoints, resulting in significant operational blind spots:

DETECTION LATENCY

14–45 Days

Average lag between illegal clear-felling and ranger physical discovery on foot patrols.

TIMBER SMUGGLING VALUE

\$150B+ / yr

Global illicit timber trade driving organized transnational smuggling cartels.

INTERCEPTION RATE

< 8.5%

Historical arrest rate when relying solely on static, uncoordinated forest checkpoints.

PUSHPA (Predictive Unified System for Forest Protection & Anti-Smuggling) overcomes these challenges by delivering an end-to-end, multi-sensor intelligence fusion platform. By marrying spaceborne optical/infrared multi-spectral imagery with ground telemetry, legal transit databases, and real-time anomaly detection, PUSHPA condenses the detection-to-interdiction loop from weeks down to minutes.

Document Table of Contents

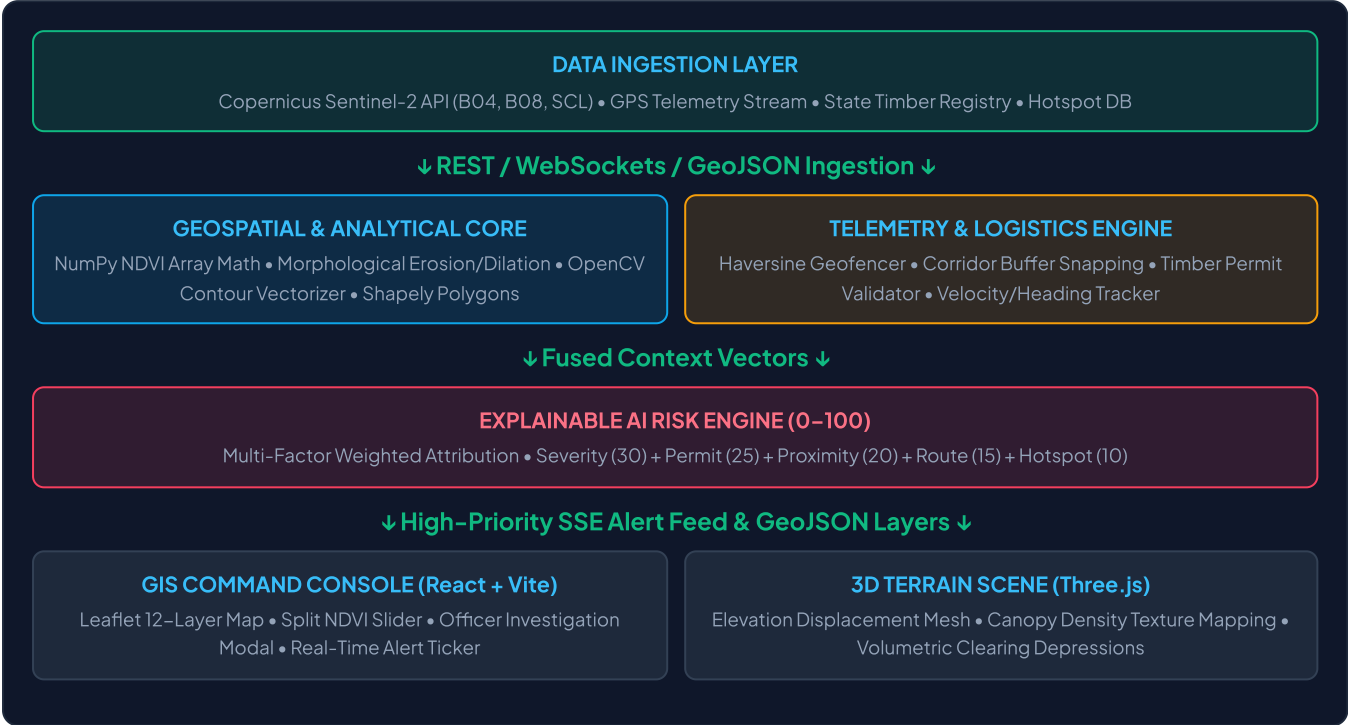
1. Executive Summary & Core Problem Statement	Page 2
2. End-to-End System Architecture & Technology Stack	Page 3
3. Copernicus Sentinel-2 Multi-Spectral Pipeline & NDVI Processing	Page 4
4. Vegetation Density Stratification & Bi-Temporal Vectorization	Page 5
5. Vehicle Telemetry, Spatial Proximity & Route Anomaly Engine	Page 6
6. Explainable AI (XAI) Investigation Risk Engine (0–100)	Page 7
7. Interactive GIS Command Dashboard & 3D Three.js Elevation Visualizer	Page 8
8. Blueprint: Real-Time Smuggling Prevention & Tactical Interception	Page 9–10
9. Database Architecture, REST API Reference & Verification Suite	Page 11
10. Ethical Governance, Legal Admissibility & Operational Deployment	Page 12

Core Architectural Philosophy

PUSHPA functions as an operational decision-support matrix. It never issues automated legal verdicts; rather, it produces transparent, factor-attributed Investigation Risk Scores and geofenced tactical evidence packets for Divisional Forest Officers (DFOs) and rapid response patrol units.

2. System Architecture & Technology Stack

PUSHPA employs a microservice-inspired, decoupled architecture dividing data ingestion, analytical tensor computation, spatial correlation, and client-side rendering into specialized tiers.



Comprehensive Technology Stack Breakdown

Subsystem Layer	Technologies Used	Architectural Role & Functionality
Backend API Core	FastAPI, Python 3.11+, Pydantic v2, Uvicorn	Asynchronous REST endpoints, auto-generating OpenAPI 3.1 docs, CORS handling, structured data validation.
Geospatial Processing	NumPy, OpenCV (cv2), SciPy, GeoPandas, Shapely	Pixel raster tensor operations, bi-temporal NDVI delta matrices, contour vector extraction, geographic centroid math.
Data Persistence	SQLite (demo), PostgreSQL + PostGIS (prod)	Spatial indexing (R-Tree / GIST), historical incident repository, timber transit permits, telemetry event logs.
Frontend Web App	React 18, TypeScript, Vite, Tailwind CSS	Ultra-responsive command UI, state-managed GIS dashboard, dark-mode styling, telemetry visualizer.
Mapping & 3D GIS	Leaflet, React-Leaflet, Three.js, WebGL	12 toggleable geospatial map layers, custom tile layers, split-screen comparison slider, 3D digital elevation model.
Alerting & Real-Time	Server-Sent Events (SSE), WebSockets	Sub-second tactical alert dispatch to officer terminals upon Critical risk threshold exceedance.

3. Copernicus Sentinel-2 Multi-Spectral Pipeline & NDVI Processing

PUSHPA integrates with the European Space Agency's (ESA) **Copernicus Sentinel-2 constellation** (Sentinel-2A & Sentinel-2B), providing multi-spectral imagery with a 5-day orbital revisit frequency and a spatial resolution of up to 10 meters per pixel.

Spectral Bands Ingested

BAND 4 (B04) — RED

665 nm

10m resolution. Absorbed by chlorophyll for photosynthesis. High reflectance indicates bare earth / cleared timber.

BAND 8 (B08) — NIR

842 nm

10m resolution. Strongly reflected by healthy leaf mesophyll cell structure. Drops sharply upon tree felling.

SCL (SCENE CLASSIFICATION)

Level-2A

Used for cloud masking, cirrus filtering, topographic cloud shadow removal, and water body isolation.

Normalized Difference Vegetation Index (NDVI) Mathematics

Live vegetation absorbs red light while reflecting near-infrared radiation. PUSHPA calculates the NDVI per pixel using normalized floating-point matrices:

$$\text{NDVI} = (\text{B08} - \text{B04}) / (\text{B08} + \text{B04} + \epsilon)$$

Where $\epsilon = 1e-6$ prevents zero-division errors over deep shadow / water surfaces.

Backend implementation from `app/satellite/ndvi.py` :

```
import numpy as np

def compute_ndvi(b04: np.ndarray, b08: np.ndarray) → np.ndarray:
    # Convert raw digital numbers to 32-bit floating point
    b04_f = b04.astype(np.float32)
    b08_f = b08.astype(np.float32)
    denominator = b08_f + b04_f
    # Apply zero-safe division mask
    ndvi = np.where(denominator ≠ 0, (b08_f - b04_f) / denominator, 0.0)
    # Clip strict physical range [-1.0, 1.0]
    return np.clip(ndvi, -1.0, 1.0)
```

Atmospheric Cloud & Shadow Masking

To eliminate false positive alerts caused by monsoonal cloud cover or terrain shadows, the pipeline applies a multi-stage mask:

- **SCL Filtering:** Pixels flagged as `3 (Cloud Shadows)`, `8 (Cloud Medium Prob)`, `9 (Cloud High Prob)`, or `10 (Thin Cirrus)` are discarded and interpolated from the nearest prior valid observation.
- **Temporal Smoothing:** Sliding-window median filters verify that a detected drop persists across consecutive satellite overpasses before triggering high-level alerts.

4. Vegetation Density Stratification & Bi-Temporal Vectorization

PUSHPA categorizes forest zones into four distinct density tiers based on calibrated radiometric thresholds, enabling rangers to identify not only total clear-cuts but also stealth degradation (thinning of mature trees under dense canopy).

Canopy Stratification Tier	NDVI Range	Ecological Characteristic	Display Palette
Non-Vegetation / Cleared	< 0.20	Bare soil, paved roads, recent clearings, rocky outcrops.	Crimson (#dc2626)
Sparse Canopy	0.20 – 0.45	Grasslands, scrub forest, severely degraded timber zones.	Amber (#f59e0b)
Moderate Canopy	0.45 – 0.70	Deciduous forest, secondary growth, plantation buffers.	Lime Green (#84cc16)
Dense / Primary Canopy	> 0.70	Virgin rainforest, high-density teak/rosewood reserves.	Deep Emerald (#059669)

Bi-Temporal Change Detection & Vectorization Algorithm

To extract actionable geographic boundaries for field rangers, PUSHPA compares baseline imagery (\$T_0\$) with current observations (\$T_1\$):

Delta NDVI Mathematical Formulation

$\Delta \text{NDVI} = \text{NDVI}_{\text{after}} - \text{NDVI}_{\text{before}}$ A pixel is flagged as **Severe Loss** if $\Delta \text{NDVI} \leq -0.30$ and $\text{NDVI}_{\text{before}} \geq 0.50$ (ensuring seasonal grass drying is not confused with mature tree felling).

1. MORPHOLOGICAL FILTERING

A 3×3 morphological kernel performs `cv2.morphologyEx(mask, cv2.MORPH_OPEN)` to remove single-pixel sensor noise, followed by dilation to bridge fragmented clear-cut clusters.

2. CONTOUR VECTORIZATION

`cv2.findContours(mask, cv2.RETR_EXTERNAL)` extracts contiguous boundaries. Contours with an area ≥ 0.5 hectares are projected to WGS84 GeoJSON polygons (e.g. `CHG_POLY_001`).

Extracted change polygon metadata attributes include:

- `polygon_id` : Unique identifier (e.g. `CHG_POLY_001`).
- `area_ha` : Geographic surface area in hectares (calculated via EPSG:3857 planar projection).
- `vegetation_drop_pct` : Mean canopy density decline within polygon bounds (e.g. `-59.2%`).
- `centroid` : Latitude / Longitude coordinates for direct GPS waypoint navigation.
- `estimated_biomass_loss_tons` : Estimated metric tonnage of felled timber calculated from canopy index.

5. Vehicle Telemetry, Spatial Proximity & Route Anomaly Engine

Satellite detection alone cannot stop timber smuggling once trees are felled. PUSHPA fuses geospatial change polygons with real-time ground transport telemetry, correlating commercial truck movements against legally gazetted timber transit corridors.

LIVE GPS TELEMETRY

1 Hz Update

Ingests Lat, Lng, Speed, Bearing, Timestamp, Cargo Weight, and Vehicle ID.

GAZETTED CORRIDORS

500m Buffer

Permitted transit corridors (e.g., NH-181, SH-8) mapped as spatial polylines with legal tolerance buffers.

ANOMALY TRIGGERS

Instant Flag

Detects off-route deviations, unmonitored dirt track access, and night curfew violations.

Haversine Spatial Proximity Formulation

The engine computes the spherical distance between moving transport vehicles and freshly vectorized clearing polygons:

$$d = 2r \cdot \arcsin(\sqrt{[\sin^2(\Delta lat/2) + \cos(lat1) \cdot \cos(lat2) \cdot \sin^2(\Delta lng/2)]})$$

Digital Timber Transit Permit Verification

Every commercial vehicle operating within forest administrative zones is cross-referenced in real-time against the state e-Forest transit registry:

Verification Check	Criteria Evaluated	Violation Consequence
Registry Existence	Vehicle registration number exists in active permit database.	UNPERMITTED_VEHICLE → Maximum penalty flag (+25 pts).
Temporal Validity	Current UTC timestamp is within valid_from and valid_to .	EXPIRED_PERMIT → Transit authorization revoked.
Declared Species Authenticity	Declared cargo matches allowed commercial quota (e.g., Teak vs. Rosewood).	SPECIES_MISMATCH → Clandestine high-value smuggling flag.
Payload Weight Tolerance	Weighbridge load vs. permitted tonnage (maximum allowable variance ±5%).	OVERLOAD_DETECTED → Unregistered timber concealment flag.
Corridor Conformance	Current vehicle coordinate lies within the designated route polyline.	OFF_ROUTE_TRANSIT → Route anomaly penalty (+15 pts).

6. Explainable AI (XAI) Investigation Risk Engine (0–100)

A critical flaw of traditional "black-box" machine learning models in law enforcement is their inability to justify why a vehicle was stopped. PUSHPA features a deterministic, transparent, and legally defensible **Investigation Risk Engine (0–100)** that produces granular factor attribution for every alert.

Weighted Risk Attribution Matrix

$$RiskScore = \min(100, W_{change} + W_{permit} + W_{proximity} + W_{route} + W_{hotspot})$$

Risk Factor	Weight	Trigger Condition & Mathematical Rule	Operational Rationale
Change Severity	+30 pts	Nearest clearing polygon exhibits canopy decline > 50%.	Distinguishes major mechanized clear-felling from minor natural branch falls.
Permit Violation	+25 pts	Vehicle is unpermitted, permit expired, or cargo mismatch.	High correlation with black-market illicit timber hauling operations.
Spatial Proximity	+20 pts	Vehicle is within ≤ 3.0 km geodesic radius of fresh clearing.	Direct spatial correlation connecting the carrier to the harvest site.
Route Anomaly	+15 pts	Vehicle departs legal highway corridor onto unpaved dirt tracks.	Smugglers bypass primary police checkpoints via unpaved forest paths.
Historical Hotspot	+10 pts	Zone contains > 5 historical illegal-felling cases within 3.0 km.	Identifies repeat poaching corridors and habitual timber mafia territory.

Investigation Risk Tiers & Tactical Protocols

0 – 34 LOW

Standard transit. Normal routine logging; no officer intervention needed.

35 – 59 MODERATE

Advisory flag. Routine document inspection at next scheduled checkpoint.

60 – 79 HIGH

Priority inspection. Mobile ranger patrol redirected to verify vehicle credentials.

80 – 100 CRITICAL

Immediate interdiction! Auto-dispatches rapid interception squad to block escape route.

Sample XAI Response Payload

```
{
  "vehicle_id": "TN01AB1234",
  "risk_score": 91,
  "rating": "CRITICAL",
  "breakdown": [
    { "factor": "Change Severity", "points": 30, "detail": "Nearest tile shows 59.2% canopy loss." },
    { "factor": "Permit Violation", "points": 25, "detail": "UNPERMITTED - No digital permit on file." },
    { "factor": "Spatial Proximity", "points": 20, "detail": "Vehicle is 0.42km from clearing CHG_POLY_001." },
    { "factor": "Route Anomaly", "points": 15, "detail": "Vehicle 1.8km off legal transit corridor." },
    { "factor": "Historical Hotspot", "points": 10, "detail": "8 prior felling incidents within 3.0km." }
  ]
}
```


7. Interactive GIS Command Dashboard & 3D Three.js Elevation Visualizer

The PUSHPA frontend is engineered as a military-grade, dark-mode GIS Command Center providing forest officers with instant spatial awareness, split-screen temporal comparisons, and immersive 3D terrain rendering.

12-LAYER GIS MAPPING SYSTEM

Integrated via Leaflet with dynamic layer controls:

- Base Cartography (Dark CartoDB, OpenStreetMap, Satellite Hybrid).
- NDVI Density Heatmap & False-Color Infrared Overlays.
- Vectorized Change Polygons (Color-coded by severity).
- Real-time Vehicle GPS Tracks & Heading Vectors.
- Legal Transit Highway Corridors & Checkpoints.
- Historical Poaching Incident Hotspot Clusters.

BEFORE / AFTER SPLIT-SCREEN SLIDER

Interactive swipe-slider allowing field investigators to visually compare optical and NDVI rasters between pre-clearing baseline (\$T_0\$) and post-clearing (\$T_1\$) satellite passes to confirm tree felling before dispatching units.

3D Terrain Elevation & Canopy Depletion Visualizer (Three.js)

To assist tactical teams in understanding steep valley topography and inaccessible ridges, PUSHPA features a custom WebGL 3D terrain viewer:

DIGITAL ELEVATION MESH

Three.js `PlaneGeometry(128, 128)` with vertex shader displacement mapping terrain ridges and ravines.

CANOPY DENSITY TEXTURE

Dynamic canvas texture mapping NDVI density gradients directly onto the 3D undulating terrain surface.

VOLUMETRIC DEPRESSIONS

Real-time wireframe depression zones highlighting clear-cut voids where tree canopy volume was extracted.

10-Step Automated Intelligence Storyboard (DEMO INCIDENT)

PUSHPA includes an automated demonstration runner executing the entire intelligence pipeline in sequence:

Steps 1-4: Earth Observation & Change Vectorization

Ingests Sentinel-2 Level-2A tiles for Nilgiri Zone A → Computes NDVI delta (-59.2% drop) → Vectorizes 2.73 ha clearing polygon `CHG_POLY_001` with exact GPS centroid.

Steps 5-8: Telemetry, Route & Permit Correlation

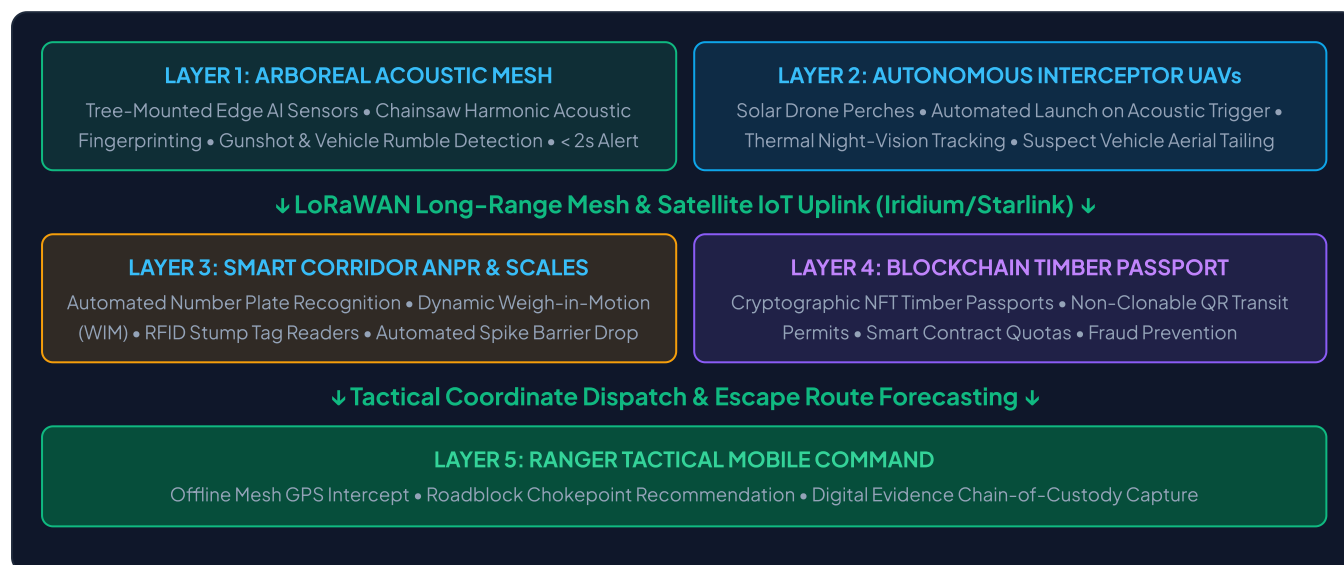
Tracks logging truck `TN01AB1234` traveling 0.42 km from clearing → Flags unpaved interior track deviation → Discovers permit is `UNPERMITTED` → Correlates with 8 prior poaching incidents.

Steps 9-10: Risk Computation & Tactical Dispatch

Risk engine computes **91/100 (CRITICAL)** → Triggers red alert ticker across all command consoles → Auto-opens the **Officer Investigation Panel (#CHG001)** with one-click tactical ranger interception dispatch.

8. Blueprint: Real-Time Smuggling Prevention Architecture

While satellite orbits (5-day revisit) detect the physical clearings, **preventing timber smuggling in real-time** requires an active, multi-layered sensor and interception grid that halts smugglers during the act of felling, loading, and transporting timber. Below is the comprehensive architectural blueprint for real-time prevention.



1. Arboreal Bio-Acoustic IoT Sensor Grid ("Forest Ears")

Solar-powered, weatherproof acoustic sensor nodes deployed in high-value tree canopies (every 500m in protected sandalwood/rosewood groves) running **Edge TinyML Audio Classifiers** (TensorFlow Lite / Edge Impulse):

- **Acoustic Fingerprint Detection:** Continuously analyzes acoustic spectrums (120 Hz – 8 kHz) to detect the two-stroke engine harmonics of chainsaws, handsaws, tree-falling shockwaves, heavy diesel engine rumbles, and gunshots.
- **Sub-Second Edge Inference:** Runs on ultra-low-power microcontrollers (Nordic nRF5340 / ESP32-S3 + MEMS microphone) consuming < 15mW, waking up only upon acoustic threshold trigger.
- **LoRaWAN Long-Range Uplink:** Transmits acoustic alert packets with precise sensor GPS ID over a 15 km forest mesh network to base stations, achieving detection-to-alert latency of **under 3 seconds**.

2. Autonomous Tactical Drone Interception Fleet (UAV First-Response)

Deploying automated drone nesting stations (solar-powered weatherproof hangars) positioned across strategic forest ridges:

- **Instant Autonomous Scramble:** When an acoustic sensor detects a chainsaw or the PUSHPA engine flags an off-route vehicle, the nearest drone launches autonomously within 45 seconds without requiring manual pilot setup.
- **Dual Optical / FLIR Thermal Payload:** Penetrates tree canopy cover at night to track thermal signatures of poachers, woodcutters, campfires, and running truck engines.
- **Automated Target Tracking & Spotlight:** Computer vision tracking locks onto fleeing vehicles, transmitting live HD video and GPS coordinates to the command center while illuminating the suspect

with a 10,000–lumen strobe.

8. Real-Time Smuggling Prevention Architecture (Cont.)

3. Smart Corridors: ANPR, Dynamic Weighbridges & RFID Stump Tags

Preventing smuggled logs from exiting forest boundaries into commercial timber markets requires automated physical choke-points:

ANPR OPTICAL GATES

High-speed ANPR cameras scan vehicle license plates at entry/exit points, querying the PUSHPA database in < 200ms. Unpermitted trucks trigger automated pneumatic spike barriers.

WEIGH-IN-MOTION (WIM)

Sensors embedded in road asphalt weigh moving trucks without stopping. Density anomalies (e.g. hollowed-out cargo concealing high-density red sanders) trigger mandatory physical inspection bays.

RFID TIMBER STUMP NAILS

Forest departments tag legally harvestable trees with tamper-evident RFID/NFC tree-stump nails. Felled logs without valid cryptographic RFID tags are automatically seized at checkpoint scanners.

4. Blockchain-Backed Timber Transit Passport & Chain-of-Custody

Paper transit permits are notoriously vulnerable to counterfeiting, reuse, and bribery. PUSHPA integrates a distributed ledger (Hyperledger Fabric / Polygon Private Enterprise):

- **Cryptographic Digital Transit Passports:** Every legal timber consignment receives a time-locked, geofenced, cryptographically signed NFT/Permit token containing harvest GPS, species genetics, log count, and destination mill.
- **Single-Use Burn Mechanism:** Once a vehicle passes the final destination checkpoint, the smart contract automatically burns the permit token, mathematically preventing permit reuse for multiple smuggling runs.
- **DNA & Isotope Timber Verification:** Logs can be cross-verified against regional wood isotopic databases to ensure claimed timber origin matches genuine provenance.

5. Graph-Based Predictive Escape Route Forecasting & Chokepoint Interdiction

When an unauthorized vehicle is detected deep inside the forest, PUSHPA's **Predictive Escape Engine** executes a Dijkstra / Graph Neural Network (GNN) traversal across the topological forest road network:

Tactical Interdiction Vector Calculation

The algorithm computes the 3 most probable exit gates based on current vehicle heading, road gradient, unpaved trail passability, and historical smuggling escape routes. It calculates the **Time-to-Exit (\$T_{\{exit\}}\$)** versus the **Ranger Interception Time (\$T_{\{ranger\}}\$)** and automatically routes patrol jeeps to the optimal bottleneck chokepoint before the smuggler arrives.

6. PUSHPA Field Command Mobile Application

OFFLINE PEER-TO-PEER MESH SYNC

In remote forest valleys without cellular reception, ranger handsets communicate over Bluetooth / LoRa mesh, sharing live suspect GPS vectors and tactical positions between squad members.

DIGITAL EVIDENCE & SEIZURE DOSSIER

On-device camera module generates cryptographically watermarked, GPS-stamped photo evidence packets of seized timber, creating court-admissible custody logs compliant with statutory evidence laws.

9. Database Architecture, REST API Reference & Verification Suite

PUSHPA exposes a standardized, high-performance RESTful API compliant with OpenAPI 3.1 specifications, enabling seamless interoperability with national wildlife crime databases, police dispatch systems, and remote field sensors.

Key REST API Endpoints

Method & Route	Parameters / Request Body	Response Payload & Description
GET /api/forests/zones	None	Returns list of monitored forest zones, boundary geometries, centroids, and baseline acreage.
GET /api/satellite/observe	zone_id, before_date, after_date	Fetches Sentinel-2 B04/B08 bands, computes NDVI rasters, and calculates mean canopy density deltas.
GET /api/geospatial/changes	zone_id, min_loss_pct, min_area_ha	Extracts and returns GeoJSON collection of vectorized clearing polygons with area and centroids.
GET /api/vehicles/telemetry	zone_id, active_only	Streams live GPS coordinates, speeds, headings, and spatial corridor proximity flags for all active vehicles.
POST /api/permits/verify	{ vehicle_id, species, cargo_weight }	Validates vehicle against digital transit registry; returns status (VALID, EXPIRED, UNPERMITTED).
POST /api/risk/evaluate	{ vehicle, change_polygons }	Computes 0–100 Explainable AI Risk Score with factor breakdown and tactical recommendations.
GET /api/demo/scenario	None	Executes the complete 10–step Nilgiri Zone A demonstration story and returns full audit trace.
GET /api/alerts/stream	auth_token	Server-Sent Events (SSE) persistent stream broadcasting live tactical alerts to officer consoles.

Database Entity Relationship Overview

ForestZone id (PK, VARCHAR) name (VARCHAR) center_lat (FLOAT) center_lng (FLOAT) area_ha (FLOAT) boundary_geom (POLYGON)	ChangePolygon polygon_id (PK, VARCHAR) zone_id (FK → ForestZone) detection_date (DATE) area_ha (FLOAT) vegetation_drop_pct (FLOAT) geom (POLYGON)	VehicleTelemetry & Permit vehicle_id (PK, VARCHAR) permit_id (VARCHAR, NULLABLE) carrier_company (VARCHAR) status (VALID / UNPERMITTED) approved_route_geom (LINESTRING) current_risk_score (INTEGER)
---	--	--

Automated Verification & Test Suite

The PUSHPA codebase includes a test suite verifying algorithmic integrity across all modules:

- test_ndvi_math : Validates zero-division handling, floating point precision, and normalization clipping $[-1.0, 1.0]$.
- test_contour_extraction : Confirms morphological filters correctly vectorize contiguous clearings while rejecting sub-threshold noise.

- `test_haversine_distance` : Verifies geodesic distance accuracy against benchmark spatial coordinates across mountainous terrain.
- `test_risk_engine_bounds` : Asserts that risk scores strictly remain bounded in $[0, 100]$ and accurately sum contributing factor points.

10. Ethical Governance, Legal Admissibility & Operational Deployment

Deploying artificial intelligence and geospatial surveillance in sensitive biosphere reserves requires rigorous adherence to ethical standards, statutory environmental laws, and tribal community rights.

1. Human-in-the-Loop Advisory Boundary

Statutory Legal Safeguard

PUSHPA is strictly an **intelligence prioritization and decision-support system**. The software NEVER autonomously declares an event illegal, issues legal fines, or executes automated punitive enforcement. All alerts are categorized as *"Potential Suspicious Activity / Investigation Required"*, ensuring that a qualified Forest Range Officer (FRO) or Divisional Forest Officer (DFO) performs physical verification before legal proceedings begin under the Wildlife Protection Act and Indian Forest Act.

2. Privacy & Indigenous Community Rights (FRA Compliance)

- **Tribal Foraging Protection:** Algorithms are calibrated to distinguish between traditional non-timber forest produce (NTFP) collection by indigenous forest dwellers (which produces negligible canopy deltas) and commercial industrial clear-felling.
- **Privacy Preservation:** Satellite optical processing occurs at 10m ground resolution, inherently preserving individual human privacy while resolving tree canopies and vehicular assets.

3. Hardware Bill of Materials (BOM) for Operational Field Pilot

Subsystem Component	Hardware Specification	Estimated Cost (USD)	Deployment Footprint
Acoustic Edge IoT Node	ESP32-S3 + MEMS Mic + 5W Solar + LoRaWAN	\$45 per node	1 node per 25 hectares in high-risk zones.
LoRa Gateway Base Station	Outdoor 8-Channel LoRaWAN + 4G/Satellite Uplink	\$380 per station	1 gateway per 15 km radius.
Autonomous Interceptor UAV	Hexacopter + Dual FLIR Thermal/4K + Solar Hangar	\$8,500 per station	1 drone station per forest range (5,000 ha).
Smart Checkpoint ANPR & Scale	Dual 8MP ANPR Cameras + Weigh-in-Motion Pad	\$4,200 per checkpoint	Installed at all gazetted forest exit highways.
Ranger Rugged Tablet	IP68 Android Handset with LoRa P2P Transceiver	\$320 per unit	Assigned to every 2-person patrol team.

Conclusion & Project Impact

PUSHPA bridges the historic technological gap between spaceborne Earth Observation and ground forest protection. By pairing automated satellite NDVI change detection with real-time acoustic IoT meshes, autonomous UAV interception, and an Explainable AI risk engine, PUSHPA equips forest guardians with a predictive, real-time defense shield to preserve our planet's most vulnerable ecosystems.

PUSHPA Technical Dossier • Predictive Unified System for Forest Protection & Anti-Smuggling

Engineered with Copernicus Sentinel-2, FastAPI, React, Leaflet & Three.js • Confidential Forest Intelligence Platform